ANSWER SCHEME
MG3DSEMATA02 – Numerical Methods for Scientific Computing

(2024 Admission Onwards)

Part A – Short Answer Questions (Any 5 × 2 = 10 Marks)

1. **Convert $2F_{16}$ to decimal.**
   $2F_{16} = 2 \times 16 + 15 = 32 + 15 = 47$.

2. **Define a numerically unstable algorithm.**
   An algorithm is *numerically unstable* if small round–off or truncation errors in input cause large errors in the output due to error amplification.

3. **One iteration of the bisection method for $f(x) = x^2 - 10$ on $[3, 4]$.**
   $f(3) = -1,\ f(4) = 6$. Midpoint $c = 3.5,\ f(3.5) = 2.25 > 0$.
   New interval: $[3, 3.5]$.

4. **Forward elimination in Gaussian elimination.**
   Row operations are used to eliminate coefficients below the main diagonal to convert the matrix into upper–triangular form; then back substitution gives the solution.

5. **LU factorization of**

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

   $L = I,\ U = A$. Hence $A = LU$ with

$$L = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

   ;

$$U = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

6. **Closed–type integration rules.**
   Integration rules that use both endpoints of the interval, e.g. Trapezoidal and Simpson's rules.

7. **Steps of Richardson extrapolation (Euler method).**
   If y_h and $y_{h/2}$ are Euler approximations (order $p = 1$),

$$y \approx y_{h/2} + \frac{y_{h/2} - y_h}{2^1 - 1} = 2y_{h/2} - y_h.$$

8. **Existence and uniqueness theorem.**
   We assume that $f(t, u)$ satisfies the following conditions:

   (i) $f(t, u)$ is a real-valued function.

   (ii) $f(t, u)$ is defined and continuous in the strip

$$t \in [t_0, b], \quad u \in (-\infty, \infty).$$

   (iii) There exists a constant L such that for any $t \in [t_0, b]$ and any u_1, u_2,

$$|f(t, u_1) - f(t, u_2)| \leq L|u_1 - u_2|,$$

   where L is called the *Lipschitz constant*.

   Then, for any initial value u_0, the initial value problem has a unique solution $u(t)$ for $t \in [t_0, b]$.

Part B – Short Essay Questions (Any 5 × 6 = 30 Marks)

(9) **Convert to binary fractions.**
(a) $0.859375_{10} = 0.110111_2$.
(b) $0.7_{10} \approx 0.1011001100110011\ldots_2$ (repeating).

(10) **Gauss–Seidel iteration for** $2x + y = 4,\ x + 3y = 5$.
Rearrange: $x = (4 - y)/2,\ y = (5 - x)/3$.
Start $(x, y) = (0, 0)$:

$$x^{(1)} = 2, \qquad y^{(1)} = 1.$$

Hence first GS iterate $\boxed{(x, y) = (2, 1)}$.

(11) **Jacobi iteration for** $4x + y = 7,\ x + 3y = 8$.
Jacobi form:

$$x^{(k+1)} = \frac{7 - y^{(k)}}{4}, \quad y^{(k+1)} = \frac{8 - x^{(k)}}{3}.$$

Start $(0, 0)$: first iteration $(1.75, 2.667)$; second $(1.083, 2.083)$.
Hence after two iterations: $\boxed{(x, y) \approx (1.083,\ 2.083)}$.

(12) **Natural cubic spline for** $x = 1, 2, 3, 4$ with $f = 0, 1, 0, 0$.
Using uniform $h = 1$ and natural boundary $M_0 = M_3 = 0$.
The tridiagonal system for M_1, M_2 gives approximate solution

$$M_1 \approx -3.6, \quad M_2 \approx 2.4.$$

Spline pieces (approximate):

$$\begin{aligned}
S_1(x) &= 0 + 1.6(x - 1) - 0.6(x - 1)^3, \quad 1 \le x \le 2, \\
S_2(x) &= 1 - 0.20(x - 2) - 1.8(x - 2)^2 + 1.0(x - 2)^3, \quad 2 \le x \le 3, \\
S_3(x) &= 0 - 0.80(x - 3) + 1.2(x - 3)^2 - 0.40(x - 3)^3, \quad 3 \le x \le 4.
\end{aligned}$$

(13) **Least squares fit to given data.**

We have

$$\sum x_i = 3, \quad \sum x_i^2 = 2.2, \quad \sum f(x_i) = 3.748, \quad \sum x_i f(x_i) = 2.5224.$$

The normal equations for fitting a straight line

$$P_1(x) = c_0 + c_1 x$$

are

$$\begin{aligned}
5c_0 + 3c_1 &= 3.748, \\
3c_0 + 2.2c_1 &= 2.5224.
\end{aligned}$$

The solution of this system is

$$c_0 = 0.3392, \quad c_1 = 0.684.$$

Hence, the required approximation is

$$P_1(x) = 0.3392 + 0.684x.$$

The least squares error is given by

$$\text{Least squares error} = \sum_{i=0}^{4} [f(x_i) - (0.3392 + 0.684x_i)]^2 = 0.00245.$$

(14) **Piecewise linear interpolation.**
Between $(1,1)$ and $(2,8)$:

$$f(1.5) = 1 + \frac{8-1}{2-1}(0.5) = \boxed{4.5.}$$

(15) **Interpolation error term.**

$$E_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

(16) **Derivative of interpolation error.**

$$E_n'(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} P_n'(x), \qquad P_n(x) = \prod_{i=0}^{n} (x - x_i).$$

Part C – Essay Questions (Any 3 × 10 = 30 Marks)

(17) **Normalized floating–point form of** -2735.46875.
Base 10: $-2735.46875 = -2.73546875 \times 10^3$.
Base 2: $2735.46875_{10} = 101010101111.01111_2 = 1.0101010111101111_2 \times 2^{11}$.
Hence normalized form: $\boxed{-1.0101010111101111_2 \times 2^{11}}$.

(18) **Newton–Raphson for** $f(x) = x^3 - 2x^2 - 5$, $x_0 = 2$.
$f'(x) = 3x^2 - 4x$. Iterations:

$$x_1 = 3.25,\ x_2 = 2.8110,\ x_3 = 2.6980,\ x_4 = 2.6907.$$

Thus $\boxed{x \approx 2.6906}$ (to 4 d.p.).

(19) **Solve linear system by matrix methods.**
Form $AX = B$, use Gaussian elimination or $X = A^{-1}B$. (Steps depend on exact matrix given; final solution to be substituted here.)

(20) **Lagrange interpolation for** $f(x) = \sin x$ on $[1,3]$.
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(21) **Runge–Kutta 4th order method**
For $y' = x^2 + y^2,\ y(0) = 0,\ h = 0.2$.
Use

$$\begin{aligned} k_1 &= f(x_n, y_n), \\ k_2 &= f(x_n + \tfrac{h}{2},\ y_n + \tfrac{hk_1}{2}), \\ k_3 &= f(x_n + \tfrac{h}{2},\ y_n + \tfrac{hk_2}{2}), \\ k_4 &= f(x_n + h,\ y_n + hk_3), \\ y_{n+1} &= y_n + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Two steps from 0 to 0.4 yield

$$\boxed{y(0.4) \approx 0.02133.}$$

(22) **Taylor series method (2nd order)**
Given $u''' + 2u'' + u' - u = \cos t$, $u(0) = 0, u'(0) = 1, u''(0) = 2$, $h = 1$.
Let $u_1 = u,\ u_2 = u',\ u_3 = u''$. Then

$$u_1' = u_2, \quad u_2' = u_3, \quad u_3' = \cos t - 2u_3 - u_2 + u_1.$$

At $t = 0$, $[u_1, u_2, u_3] = [0, 1, 2]$. Then

$$y'(0) = [1, 2, -4], \quad y''(0) = [2, -4, 7].$$

Second–order Taylor step ($h = 1$):

$$y(1) = y(0) + y'(0) + \frac{1}{2}y''(0) = [2.0,\ 1.0,\ 1.5].$$

Hence

$$u(1) = 2.0, \quad u'(1) = 1.0, \quad u''(1) = 1.5.$$